



0	1	2	3	4	5	6	7
1	1	2	10	4	5	2	1

2IS



1	2	2	3	3	4	2	1
---	---	---	---	---	---	---	---

2IS

1	5	2	4	3	3	2	1
---	---	---	---	---	---	---	---

Consider

$$2 + 5 \Rightarrow 7 \cdot 1 \neq \textcircled{6}$$

$$4 \cdot 1 \neq 3 \rightarrow$$

Binary Search $\rightarrow$ LIS $\rightarrow n \log n \Rightarrow$

2min

0	1	2	3	4	5	6	7	8	9	10
10	8	4	9	6	1	5	8	7	6	12

LIS

0	1	2	3	4	5	6	7	8	9	10
1	1	1	2	2	1	2	3	3	3	1

1

10	8	9	1
----	---	---	---

chota hga $\rightarrow$ leny 2

1	5	6
---	---	---

2

9	6	5
---	---	---

3

8	7	6
---	---	---

109n

